HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information Communication Technology

GCP-ready

Mobile + desktop

UC001–UC005

CAPSTONE PRESENTATION

Ecopark Bicycle Parking Platform

Ecopark bicycle rental and return on Google Cloud Platform

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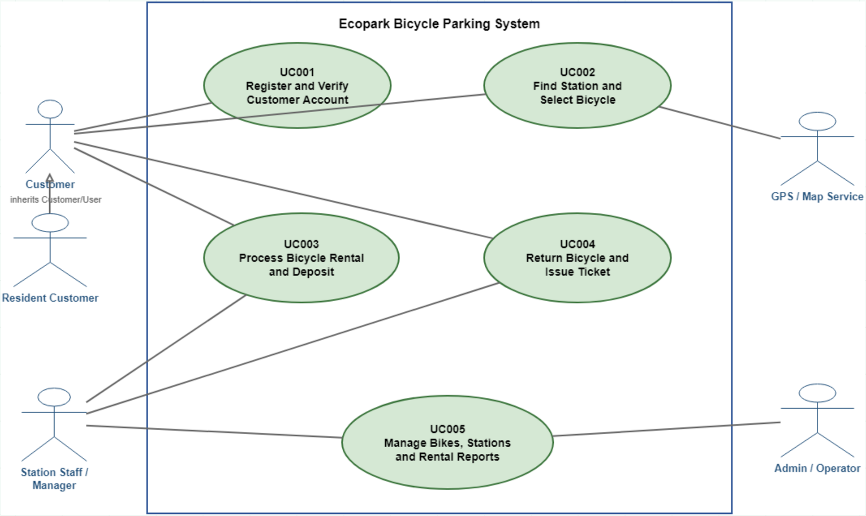
Problem and Objectives

- Ecopark has multiple stations and bicycle types whose availability changes continuously with each rental.
- Customers need to find nearby stations, select available bicycles, submit requests, and track rental status.
- Staff and administrators must verify documents, collect a VND 200,000 deposit, hand over bicycles, accept cross-station returns, and issue tickets.
- Managers need revenue dashboards, fleet status, audit logs, and filtered CSV reports.
- The team built a responsive web platform for desktop and mobile, ready for deployment on GCP.

Actors and Use Cases



Implemented Scope: UC001–UC005

Use Case	Objective	Platform Outcome
UC001	Customer accounts and verification	Registration and login; validation for name, phone number, ID document, and resident discount.
UC002	Find stations and select bicycles	Live or manual GPS, distance and type filters, bicycle selection, request submission/cancellation, and handover geofence.
UC003	Handover and deposit	Staff verify documents, collect a VND 200,000 deposit, retain documents, and mark the bicycle as rented.
UC004	Return and ticket issuance	Cross-station return, fee calculation, resident discount, late surcharge, condition notes, and ticket summary.
UC005	Operations management	Station and fleet CRUD, status reasons, report charts, audit logs, and CSV export.

ECOBIKE DEPLOYMENT AND RUNTIME STACK

EcoBike deployment architecture from users to data

ecobike.ccat.io.vn
HTTPS demo endpoint via Cloudflare Tunnel

1 User devices
Customers, staff, admins, and the GPS tab access the same EcoBike platform.
Desktop · Mobile · /gd demo

2 Cloudflare DNS + Tunnel
Public HTTPS routing reaches the VM without exposing the app through an inbound firewall rule.
Cloudflare · DNS · Tunnel · No inbound firewall

3 GCP Compute Engine free tier
The live demo VM keeps the app and tunnel running through systemd or Docker restart policies.
Google Cloud · Compute Engine · Ubuntu VM · Persistent disk

4 Dockerized Node.js service
The container packages the static UI and HTTP API, bound internally to localhost:4173.
Docker · Node.js · PORT 8080 · Restart unless-stopped · Single image

5 App runtime
The browser runs the vanilla JavaScript UI, motion, 3D scene, and interactive map.
JavaScript · GSAP · Three.js · Leaflet · Responsive UI · 3D hero · Map flow

6 Backend modules
Business modules handle authentication, requests, handover, return tickets, reports, and audit logs.
HTTP API · Auth · Rental · Handover · Ticket · Report · Audit

7 Data layer
SQLite/WAL persists demo data; Cloud SQL is the scale-up path.
SQLite · Future Cloud SQL · Users · Bikes · Rentals · Logs

EcoBike Ecopark Bicycle Rental Platform

Docker on GCP VM + Cloudflare Tunnel + SQLite persistent volume

Data Design

Pricing policy
resident discount, late surcharge, deposit

Users & profiles
USERS
CUSTOMER_PROFILES
RESIDENT_CARDS
STAFF

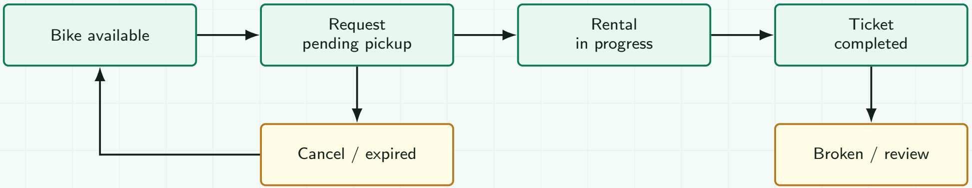
Rental transactions
RENTAL_REQUESTS
RENTALS
DEPOSITS
TICKETS

Stations & fleet
STATIONS
BIKE_TYPES
BIKES
BIKE_STATUS_LOGS

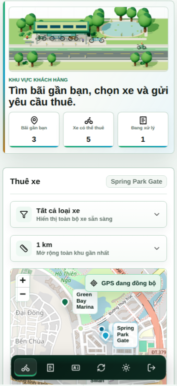
Operations reporting
revenue, fleet status, audit trail

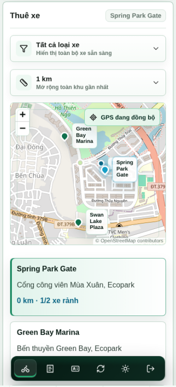
The detailed ERD is in the report; this slide shows only the core data relationships around the rental lifecycle.

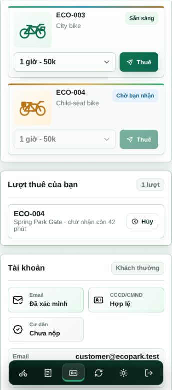
Bicycle Rental Lifecycle



Customer Mobile Flow



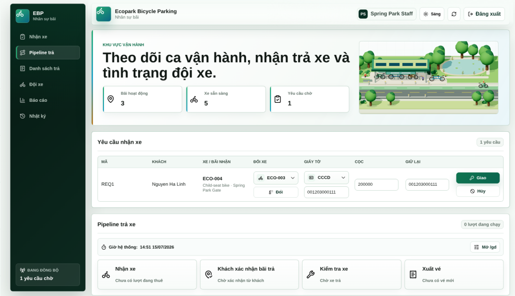


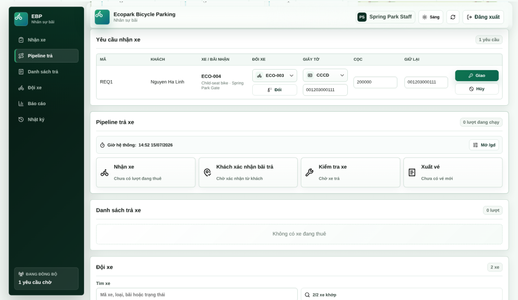


Overview	Find a station	Select a bicycle

The mobile flow is split into unobstructed views above the bottom dock: overview, nearest-station results, and available-bicycle selection.

Operations UI



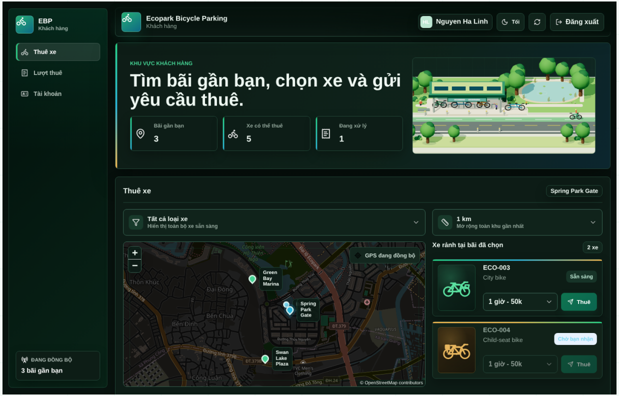


Operations overview

Pickup requests and return pipeline

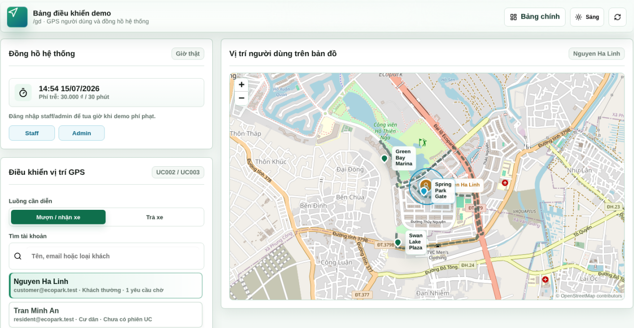
Operations screenshots capture each state separately so the overview and processing areas remain clear during the presentation.

Light/Dark Mode



- Light, dark, and system themes are persisted in the browser.
- The Leaflet map has a dedicated dark treatment that avoids a harsh white map surface.
- Account cards, dropdowns, charts, and forms share the same design tokens.
- The 3D scene preserves enough contrast to distinguish bicycles, roads, the lake, and trees.

Demo Director for Alternative Flows



1. **Select a user / session**
   Find a customer account and select its session when available.

2. **Move the user's GPS**
   Stream the route polyline to the customer map.

3. **Fast-forward system time**
   Demonstrate expiration and late surcharges.

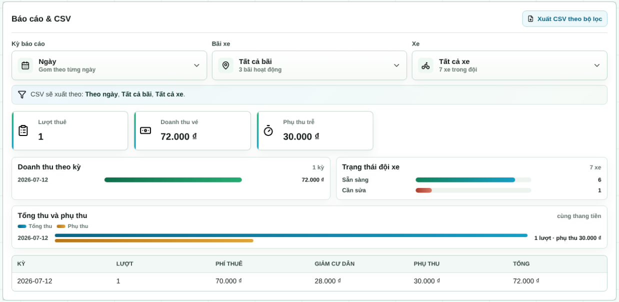
4. **Validate the station radius**
   Requests may be created anywhere; handover and return require proximity.

API and Backend Implementation

API Group	Role	Main Endpoints
Auth/Profile	Public/Customer	`/api/auth/register`, `/api/auth/login`, `/api/customer/resident-card`.
Station/Bike	Customer	`/api/stations`, `/api/stations/:id/bikes`, `/api/rental-requests`.
Handover/Return	Staff/Admin	`/api/staff/pending-requests`, `/api/staff/handover`, `/api/staff/return`.
Admin/Report	Staff/Admin	`/api/admin/bikes`, `/api/admin/reports`, `/api/admin/bike-status-logs`.

- Server-side validation covers names, phone numbers, ID documents, and duplicate data.
- Rental, handover, and return operations use transactions to prevent double-booking.
- Audit logs record bicycle status changes together with the operator, station, and reason.

Reporting Dashboard



- Revenue charts use a consistent monetary scale across reporting periods.
- Fleet status distinguishes available, rented, and maintenance-required bicycles.
- Rental counts and late surcharges appear alongside the underlying report table.
- CSV exports preserve the filters currently selected on the dashboard.
- Audit logs identify who changed each bicycle's status and why.

Testing and Evidence

24/24
Backend API tests
Station staff + rental flow

5
UI viewports
Customer, staff, admin

UC
End-to-end smoke
/gd, handover, return, CSV

CI/CD
GitHub Actions
Check, coverage, smoke, Docker

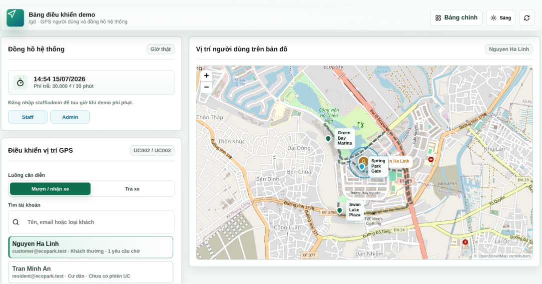
80.67%
Line coverage
Node.js built-in coverage

71.46%
Branch coverage
Node.js built-in coverage

84.92%
Function coverage
Node.js built-in coverage

Test scenario: open /gd, select the correct user, verify that the customer marker follows the GPS stream, submit a rental request, complete staff handover, confirm a return station within the valid radius, issue the ticket, and inspect the dashboard, CSV export, and audit log.

Open the Live Demo



1. Open the Customer, Staff/Admin, and /gd tabs.
2. Submit a customer request, then use /gd to move the user into pickup range before handover.
3. Staff verify documents, collect the VND 200,000 deposit, and hand over the bicycle.
4. The customer confirms the return station; staff issue the ticket and review any surcharge.
5. Finish with the dashboard, audit log, and CSV export.

Live demo: complete all five steps in the app.